Week 10 LOGISTIC REGRESSION

Data Preparation

Set the working directory to the folder where you will be saving the data files and the output files from the analysis exercises. In the output file name it is a .txt file, which can be opened in any text editor.
Download the data file from BB and place it into that same directory.

```
setwd("C:/Data")
options(width=80)
sink("week10.out")
dim(mydata <- read.csv("hsb2.csv"))
mydata$id <- factor(mydata$id)
mydata$female <- factor(mydata$female)
mydata$ses <- factor(mydata$ses)
mydata$schtyp <- factor(mydata$schtyp)
mydata$prog <- factor(mydata$prog)
mydata$honcomp <- ifelse(mydata$honcomp >= 60, 1, 0)
mydata$honcomp <- factor(mydata$honcomp)
dim(mydata)
```

```
[1] 200  11
```

```
summary(mydata)
```

```
       id     female     ses      schtyp     prog      honcomp     read
  70  :  1   0: 91   1: 47   1: 168   1: 45   0: 154   Min.   : 4.00
  71  :  1   1:109   2: 95   2:  32   2: 105   1:  46   1st Qu.:44.00
  72  :  1           3: 58                    3:  50   Median :54.00
  73  :  1                                            Mean   :52.23
  74  :  1                                            3rd Qu.:60.00
  75  :  1                                            Max.   :76.00
  (Other):194
     write           math           science          socst
 Min.   : 6.00   Min.   : 33.00   Min.   : 26.00   Min.   : 26.00
 1st Qu.:44.00   1st Qu.:45.00   1st Qu.:44.00   1st Qu.:41.00
 Median :54.00   Median :52.00   Median :53.00   Median :52.00
 Mean   :52.77   Mean   :52.65   Mean   :52.65   Mean   :52.41
 3rd Qu.:60.00   3rd Qu.:60.00   3rd Qu.:60.00   3rd Qu.:61.00
 Max.   :67.00   Max.   :75.00   Max.   :74.00   Max.   :71.00
```

```
table(mydata$honcomp, mydata$prog)
```

```
     1  2  3
  0 37 87 30
  1  8 18 20
```

```
table(mydata$honcomp, mydata$ses)
```

```
     1  2  3
  0 32 78 44
  1 15 17 14
```

```
table(mydata$honcomp, mydata$schtyp)
```

```
     1  2
  0 147 32
  1  21 14
```

```
table(mydata$honcomp, mydata$female)
```

```
     0  1
  0 77 77
  1 14 32
```

```
table(mydata$honcomp, mydata$id)
```

```
     70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200
  0  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1  1   1   1   
```